RB-002 — On-device voice for the character device

Name wake-word + privacy-safe trainee speech-to-text for the VRAI Faces tablet

Brief	RB-002
Status	Open — to be executed in Claude Cowork (deep research), then go/no-go → ADR + drop-in.
Gates	Heavy / maturing in-browser on-device ASR; must not breach local-first (ADR-0001) or the PHI guardrail (ADR-0014).
Decisions ratified	Name-trigger = on-device only; PTT = trainee→character (per the operator's request).
Drives	The "Character devices" workstream (ADR-0024): per-character QR device interface + hold-to-talk + name activation.

1. Objective

Let a bedside tablet running the per-character Chrome interface (avatar or audio-only) (a) **activate when the character's name is spoken nearby** ("Hey, Amy…") and (b) let a trainee **push-to-talk** to the character — with **all microphone audio processed on the device**. The transcribed text routes to the existing character-AI turn path, which returns a `VRAISpeechFrame` the avatar speaks (Phase 4.3). The voice capture/recognition is the only missing, gated piece.

2. Why this is gated

- **Privacy / PHI.** A wake-word means a continuously-listening mic. The trivial route — the browser `SpeechRecognition` (Web Speech API) — streams audio to Google's servers. Trainee speech is **trainee_input → PHI** (ADR-0014, BAA-only) and ambient capture leaving the device breaks local-first (ADR-0001). So cloud ASR is disqualified for this surface.
- **Maturity / weight.** Real on-device ASR/keyword-spotting in the browser is heavy (model download, WASM/WebGPU compute) and fast-moving. It needs measurement before we commit a dependency + a model to the tools sheet (project rule: new lib/model ⇒ ADR + tools-sheet line).

3. Current state

- Per-character **device QR** ships today (`/qr/face/<id>.svg`) → opens the interface (real face if a skin is assigned, else placeholder/head-proxy).
- The **reply loop already exists**: text → character AI turn → `VRAISpeechFrame` → on-device Kokoro TTS + lip-sync (Phases 2–4). Only speech→text on the device is missing.
- The **operator** PTT uses cloud Web Speech (operator-only, not trainee PHI) — not reusable for the bedside trainee device.

4. Research questions

1. **On-device STT engine.** transformers.js Whisper (tiny/base, WASM + WebGPU) vs. a dedicated streaming ASR — accuracy on short clinical utterances, first-token + full-utterance latency on an iPad-class tablet, model size, cold-load time, battery/thermal over a session.
2. **Name wake-word.** A lightweight always-on keyword-spotter (e.g. openWakeWord / a small CNN/RNN on mel-spectrograms) vs. running tiny-STT in a rolling buffer and matching the name. False-accept / false-reject rate for

arbitrary character names (the name is data, not a fixed model) in a noisy room.

3. **Custom name handling.** Names are per-scenario (not bakeable into a fixed wake model) — phonetic matching, fuzzy text match on rolling STT, or per-name enrollment?
4. **Chrome on-device path.** Is Chrome's on-device `SpeechRecognition` (`processLocally` , no network) available + reliable enough on the target OS/version to use directly?
5. **Capacitor/native.** Does the iOS/Android shell (ADR-0006) offer a better native on-device ASR than the web layer, and is the bridge worth it?

5. Evaluation criteria

Criterion	Target / threshold	Why
Audio leaves device	**Never** (hard gate)	ADR-0001 / ADR-0014
Wake-word false-accept	< 1 / 10 min idle	No spurious activations mid-scenario
Wake-word detect latency	< 500 ms	Feels responsive when named
PTT utterance → text	< 1.5 s after release	Keeps the turn conversational (§5 budgets)
STT word error rate	< 12% on clinical phrases	Reliable enough to drive the AI turn
Model footprint / cold load	≤ ~80 MB; ≤ ~5 s	Tablet memory + first-launch UX
Sustained cost	No thermal throttle over 20 min	Tablet runs a full scenario

6. Deliverables

- Recommended engine(s) for (a) wake-word and (b) PTT STT, with the measured numbers above.
- A pinned model + size + license, ready for a tools-sheet line, and the per-name strategy.
- A drop-in module contract: `name_trigger` (emits "addressed") + `device_stt` (PTT audio → text), both on-device, fail-closed.
- The portal trainee-input endpoint shape (text in → character AI turn → `VRAISpeechFrame`), including auth/cost handling for the no-auth device origin.

7. How the result re-enters the build

1. Go/no-go from the measured criteria → convert this brief into an ADR (engine + model pinned).
2. Add the model to `VRAI_Faces_Tools_Resources.xlsx` + `setup:assets` (lazy, never at boot).
3. Implement `name_trigger` + `device_stt` shell modules; wire PTT → portal trainee-input → character AI → speech frame (the avatar already speaks frames).
4. Ship behind the device interface that already loads via the per-character QR.

8. Hard constraints

- Microphone audio is processed **on the device**; raw audio never leaves it. ADR-0001 / ADR-0014
- Wake-word is opt-in per device and clearly indicated when the mic is live; PTT is explicit (press-and-hold).
- Recognition fails **closed** (no detection ⇒ no action); never auto-sends trainee audio anywhere.

- No new runtime dependency without an ADR + a tools-sheet line. Models load lazily, never at boot.